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Comparisons of new body load and metabolic power average workload indices between starters and non-starters: A full-season study in professional soccer players

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Abstract

The aim of this study was to compare the weekly average training monotony new body load (wTM_{NBL}) and strain (wTS_{NBL}), as well as the weekly average training monotony metabolic power average (wTM_{MPA}) and strain (wTS_{MPA}) between four periods of a season (preseason, early-season, mid-season, and end-season), considering starters and non-starters. Twenty-one professional soccer players (age: 28.27 ± 3.78 years) were monitored throughout a season in the highest level of professional football Premier League in Iran. Data were captured by Global Positioning System (GPS) devices. Independent samples *T*-tests were applied to analyze the between-group differences for all dependent derived-GPS variables for the full season and its different periods (preseason, early-season, mid-season, and end-season). Based on the amount of time attending in match and training, players were divided into two groups (starters and non-starters) each week. The magnitude of the between-group difference revealed a very large significant greater weekly average TM_{NBL} ($p < 0.001$, $d = -2.42$), TS_{NBL} ($p < 0.001$; $d = -2.74$), TM_{MPA} ($p < 0.001$; $d = -2.79$) and TS_{MPA} ($p < 0.001$; $d = -3.27$) for starters when compared to non-starters during the early-season. The findings also revealed a very large significant difference when starters were compared to non-starters during the mid-season (TM_{NBL}: $p < 0.001$, $d = -2.89$; TS_{NBL}: $p < 0.001$, $d = -2.99$; TM_{MPA}: $p < 0.001$, $d = -3.28$; and TS_{MPA}: $p < 0.001$, $d = -3.25$) and end-season (TM_{NBL}: $p < 0.001$, $d = -2.89$; TS_{NBL}: $p < 0.001$, $d = -3.07$; TM_{MPA}: $p < 0.001$, $d = -3.16$; and TS_{MPA}: $p < 0.001$, $d = -3.58$). In summary, the results of this study revealed that starters present regularly higher values of NBL, MPA-based weekly training monotony, and training strain than non-starters. This result must be taken into account when planning weekly workloads for these groups. Specifically, starters might experience high values of external workloads because of match-related demands. Therefore, weekly adjustments in their training workload are required to reduce injury risk.

Keywords

Association football, performance, sports training, training monotony, training strain

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Introduction

The investigation of soccer athletes' physical performance is commonly conducted at the elite level.[1] This performance may be improved by a training process in which workloads are regularly imposed on players, generating positive adaptations.[2,3] So, monitoring the training workload is a determinant process in elite sports, since the information collected from athletes' responses to the imposed workloads must be taken into account when planning subsequent training sessions.[4,5] Training workload (TL) can be monitored utilizing

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internal workload, which reflects the acute biological responses to a given physical stimulus during training and competition, as well as external workload, which is the measure of work performed by an athlete during training sessions and competitions.[2,6] Previous studies have shown that internal and external load may vary throughout a season in different team sports.[7,8] Sometimes this is planned by the coaching staff (e.g. in specific moments of the preseason) and other times is a result of contextual factors, such as a congested fixture scenario. Specifically, regarding variations in externally imposed workloads, it has been shown that matches induce significant biochemical and neuromuscular responses related to fatigue[9] because of the high-intensity actions demanded.[2,10] As such, matches are expected to have a significant effect on athletes' weekly workloads. This is likely to create an imbalance in the weekly training workload between starters and non-starters within a team, demanding individual adjustments in the training workloads. However, to the best of the authors' knowledge, few data are available in the literature regarding the weekly TL of starters and non-starters throughout an entire season, which impacts the available knowledge on this topic and reduces the quality of evidence-based training prescriptions.

The external workload imposed by official matches is often high because of the large amount of high-intensity activity demands required, such as accelerations and high-speed running.[2,10] With few substitutions allowed in official matches (currently only three are allowed), athletes who start a match are expected to present a higher weekly workload than non-starters. Regarding this topic, the literature has shown that substitute players perform more bouts of high-intensity running compared to other players in the second half when the data is relativized by the effective playing time.[11,12] However, since the playing time of substitute players is usually reduced, regular players are expected to present a higher external workload when considering the entire match. Also, professional soccer is currently characterized by reduced time for training sessions due to the large number of official matches. For this reason, it has been suggested that official games make the most significant contribution to the maintenance of athletes' physical fitness levels.[13] Therefore, monitoring the external workloads of starters and non-starters seems necessary to reduce injury risk to athletes who start most games, as well as to create training strategies to increase the weekly workload of non-starters, thereby avoiding a decline in their physical conditioning. However, no data is available in the literature regarding the possible differences of the external workloads of starters and non-starters in professional soccer.

Recently derived variables, such as the New Player Load (NPL) (also known as player load or body load, depending on the manufacturer), were proposed to investigate changes of direction along all the axes and can provide useful information for comprehending the external workload of training sessions and matches.[2] The external workload measure provided by the new body load (NBL) is obtained by calculating the sum of the accelerations in the three movement axis, although some differences regarding this calculation are noticeable in the literature.[14] Discriminating between constant-speed locomotion and the rate of change of velocity (i.e. acceleration) is important due to the greater energy cost and muscular demand of such movements.[15] So, acceleration demands are likely to be related to players' external workload and, consequently, must be continuously monitored. Recently developed GPS devices present an embedded triaxial accelerometer, which enables the reliable collection of players' changes in speed during game-based tasks and can be adopted to assess player load (PL) during training sessions and matches. In addition, the average metabolic power (MBA) allows for the measurement of energy expenditure, so it is an indicator of the average work intensity.[16,17] Together, the NBL and the MBA might be used by coaches to constantly monitor players' responses to workloads and make appropriate adjustments when required (e.g. increasing or reducing the training workloads).

The above mentioned variables might be used to estimate the weekly workload of the athetes when considering their fluctuation over time and varation within a week. For this purpose, training monotony and training strain have been proposed and discussed in the literature.[3,7,18] Weekly training monotony is defined as the mean of TL during the week, divided by the standard deviation (SD) of the TL of the same period.[19] This measure represents the variability of practices for the entire season.[20] Weekly training strain is defined as the sum of the TLs for all training sessions and matches during a week, multiplied by training monotony.[19] This measure reflects the overall stress demanded of the athlete during the week.[20] The literature has suggested that appropriate training variations are important to avoid high monotony and very-high relative workload, thereby minimizing injuries.[21] So, achieving an optimal very-high relative workload without increasing monotony is an important goal for strength and conditioning coaches. Considering that accelerations are relevant for understanding the external workload, it seems relevant to propose an assessment of training monotony and strain by using NBL as the input variable. However, no previous study has analyzed changes in the training monotony and strain by considering acceleration as the dependent variable for measuring the imposed workloads throughout a season.

The literature has shown that a player's effort varies throughout a season.[22] However, it is not clear if these changes are related to the player's status as a starter or non-starter. By knowing whether starters and non-starters have differences in their weekly workloads, coaches may plan specific training programs, which could promote positive adaptations and reduce the injury risk. Based on the above mentioned rationale, this study aimed to analyze the weekly training monotony and

Table 1. Training, match and total time during the periods of the season.

	Preseason	Early-season	Mid-season	End-season
Weeks (*n*)	5	14	15	14
Training sessions (*n*)	27	64	57	52
Matches (*n*)	1	15	15	13
Total time* (min)	1805.4 ± 602.7	4636.7 ± 1314.2	4014.0 ± 1418.3	3670.6 ± 1353.2

*Total time = The average of the total duration every week + The whole weeks of that section (include the total time of sessions + matches).

strain among professional soccer athletes throughout an entire season and compare weekly workload between starters and non-starters in the different periods of the season. The expectation was that the mid-season and end-season periods would present the highest monotony based on the commonly observed congested fixture,[23] while the preseason would present the most strain because of the need to recondition the athletes. Morever, starters were expected to regularly present higher training monotony and strain than non-starters.

Methods

Experimental approach to the problem

This study includes a professional team that participated in the highest level of professional football, the Premier League in Iran (Persian Gulf Premier League and knockout tournament in this country). In this league, each team is allowed to use the Global Positioning System (GPS) to record the physical fitness statistics of its players. A full season, which is 48 weeks long, was divided into four periods: preseason (weeks 1– 5); early-season (weeks 6–19); mid-season (weeks 20–34); and end-season (weeks 35–48) (Table 1). These periods were used to analyze the differences between starter and non-starter players weekly averages. The criterion to defined starters and non-starters was the player's play time during the match for each week. The players were divided into starters ($n = 10$) and non-starters ($n = 11$). Starters were defined as players who participated at least 60 min during the competition match of the week, while players who played less than 60 min, did not play at all or were not considered for the match were defined as non-starters. During each session, players were monitored by a GPS (GPSPORTS systems Pty Ltd, Model: SPI High-Performance Unit (HPU); Australian) and the study variables were collected daily during the full season (i.e. all training and matches). This study aimed to (i) describe week by week the average training monotony and strain in a NBL and MPA between starters and non-starters; (ii) and analyze the differences between starters and non-starterson the weekly average training monotony (wTM_{NBL}) and strain (wTS_{NBL}) and the weekly average training monotony (wTM_{MPA}) and strain (wTS_{MPA}) variations during four periods of the season (pre-, early-, mid-, and end-season).

Participants

Twenty-one professional soccer players (age, 28.3 ± 3.8 years; height, 181.2 ± 7.1 cm; body mass, 74.5 ± 7.7 kg; BMI, 22.6 ± 1.0 kg/m^2) participated in this study. The inclusion criterion required that players must participate in at least three training sessions each week. The exclusion criteria included: (i) players with prolonged injury or a lack of participation in training for at least two consecutive weeks (two players were removed based on this criterion); (ii) goal keepers were excluded from the study due to differences in training activities and workload in training and matches. The experimental approach and study design were presented to the players after which written consent was obtained from all players. The study followed the ethical guidelines of Helsinki Declaration for the study in Humans and was approved by the Ethics Committee of the University of Isfahan.

Monitoring external workload

GPS receiver specifications. During the season, all workouts and match sessions were monitored using GPSPORTS systems Pty Ltd. GPS-based tracking systems for professional athletes, model SPI HPU, include 15 Hz position GPS and data source NBL used a Tri-Axil accelerometer. According to a previous study, this device has a high validity and reliability.[24] There were no reported adverse weather conditions to affect data collection.

Data collection. Prior to the start of the traning session, belts were placed on the players' shoulder and chest. After each cool down session at the end of the training, the belts were collected from the players. All belts were checked by the team's GPS manager and then entered into the dock system to download the information, which was then stored on the computer with the Team AMS software. The data from each session was automatically deleted from the belt memory after download. Prior to the next session, the belts were placed in an electric charge station. The SPI IQ Absolutes were adjusted for GPS default zone throughout the season. Also, the personal characteristics (such as height and weight) of each player were entered in the software and each player registered a belt in his own name for use until the end of the season.

Measures of training workload. In this study, the evaluated variables were as follows: (1) Duration in minutes; (2) NBL obtained according to the GPS manufacturer's manual based on accelerometer data; (3) MPA obtained according to the GPS manufacturer's manual based on an individual's running speed, acceleration, and deceleration data, which shows the amount of energy consumed by the player per second (W/kg). A very good inter-reliability (3%–5%) was recorded for MPA.

The NBL was used as a training load. Total NBL in 1 week was considered as a weekly acute training load.

Then, wTM_{NBL} was calculated (weekly acute training load/SD)[25] and wTS_{NBL} was calculated (weekly acute training load $\times$ wTM_{NBL}).

For MPA, the total for a given week was considered as a weekly acute MPA. Then wTM_{MPA} was calculated (weekly acute MPA/SD) and wTS_{MPA} was calculated (weekly acute MPA $\times$ wTS_{NBL}).

Statistical analysis

Statistical analyzes and computations were carried out using the Statistical Package for the Social Sciences (SPSS, version 25.0; IBM SPSS Inc., Chicago, IL). Descriptives statistics are presented as mean (SD). The normality and homogeneity of data were tested by applying the Shapiro-Wilks and Levene's tests, respectively. Then, inferential tests were performed. Between-group differences were analyzed using the independent samples *T*-test for all dependent derived-GPS variables for the different periods of the season (preseason, early-season, mid-season and end-season). Cohen's *d* effect size (95% confidence interval) was calculated using a specific Excel spreadsheet designed for these computations. The thresholds for Cohen's *d* effect size were applied as follows: $\leqslant 0.2$, trivial; 0.2–0.6, small; 0.6–1.2, moderate; 1.2–2.0, large; 2.0–4.0, very large; and > 4.0, nearly perfect.[26] Significant level was set at $p \leqslant 0.05$.

Results

Figure 1 shows an overall vision of the weekly average training monotony (TM_{NBL}) and strain (TS_{NBL}) variations, based on new body load data, across the full season and its different periods (preseason, early-season, mid-season and end-season) for starter and non-starter players. Overall, the highest TM_{NBL} were found in week 2 and week 34 for starters and non-starters, respectively, while the lowest TM_{NBL} occurred in week 30 and week 46, for starters and non-starters, respectively (Figure 1(a)). Both groups showed the highest TS_{NBL} in week 2 and the lowest TS_{NBL} in week 46 (Figure 1(b)).

Figure 2 shows an overall vision of the weekly average training monotony (TM_{MPA}) and strain (TS_{MPA}) variations, based on metabolic power average data, across the full season and its different periods for starters and non-starters. Overall, the highest TM_{MPA} happened in week 2 for both groups, while the lowest TM_{MPA} was found in week 30 and week 46 for starters and non-starters, respectively (Figure 2(a)). Both groups presented the highest TS_{MPA} in week 2; however, the lowest TS_{MPA} occurred in week 1 and week 29 for starters and non-starters, respectively (Figure 2(b)).

Between-group comparisons on weekly average TM_{NBL}, TS_{NBL}, TM_{MPA}, and TS_{MPA} for the different periods of the season are displayed in Table 2. Coincidentally, significant differences were found in favor of starters compared to non-starters in all dependent variables during preseason (TM_{NBL}: $p < 0.001$, $d = -1.87$; TS_{NBL}: $p = 0.005$, $d = -1.32$; TM_{MPA}: $p < 0.001$, $d = -2.07$; and TS_{MPA}: $p < 0.001$, $d = -2.07$); early-season (TM_{NBL}: $p < 0.001$, $d = -2.42$; TS_{NBL}: $p < 0.001$, $d = -2.74$; TM_{MPA}: $p < 0.001$, $d = -2.79$; and TS_{MPA}: $p < 0.001$, $d = -3.27$); mid-season (TM_{NBL}: $p < 0.001$, $d = -2.89$; TS_{NBL}: $p < 0.001$, $d = -2.99$; TM_{MPA}: $p < 0.001$, $d = -3.28$; and TS_{MPA}: $p < 0.001$, $d = -3.25$); and end-season (TM_{NBL}: $p < 0.001$, $d = -2.89$; TS_{NBL}: $p < 0.001$, $d = -3.07$; TM_{MPA}: $p < 0.001$, $d = -3.16$; and TS_{MPA}: $p < 0.001$, $d = -3.58$).

Discussion

This study aimed to analyze the weekly training monotony and strain related to the external workload imposed on professional soccer players throughout an entire season and compare these workloads between starters and non-starters. In general, the results indicated small variations in monotony throughout the season, and the highest training strain was reported during the preseason, which is in line with the suggested hypotheses. Moreover, starters showed higher values of monotony and strain than non-starters in all moments of the season, and the differences were highest in the mid-season, which is also in line with the proposed hypotheses.

It has been shown in the literature that teams require greater effort (total distance, high-intensity activity, and sprints) at the end of the season in comparison to the middle of the season.[22] Also, starters accumulated greater perceived TLs than non-starters.[27] Therefore, the current results regarding the reported training strain are in line with the available literature. Starters usually exhibit a longer effective playing time than non-starters, so it is expected that they present a higher TL. This rationale was confirmed in the present study, considering that the magnitude of between-group differences was higher in the season (early, mid, and end) when compared to the preseason. Also, the highest between-groups differences in training strain were observed in the mid-season, which reinforces the importance of official matches and their accumulation for the players' weekly workloads.[2,10,27] However, significant between-group differences were also observed in the preseason for wTS_{NBL} and wTS_{MPA} when only one match was played. For this reason, a priori individual differences in players' physical skills between starters and non-

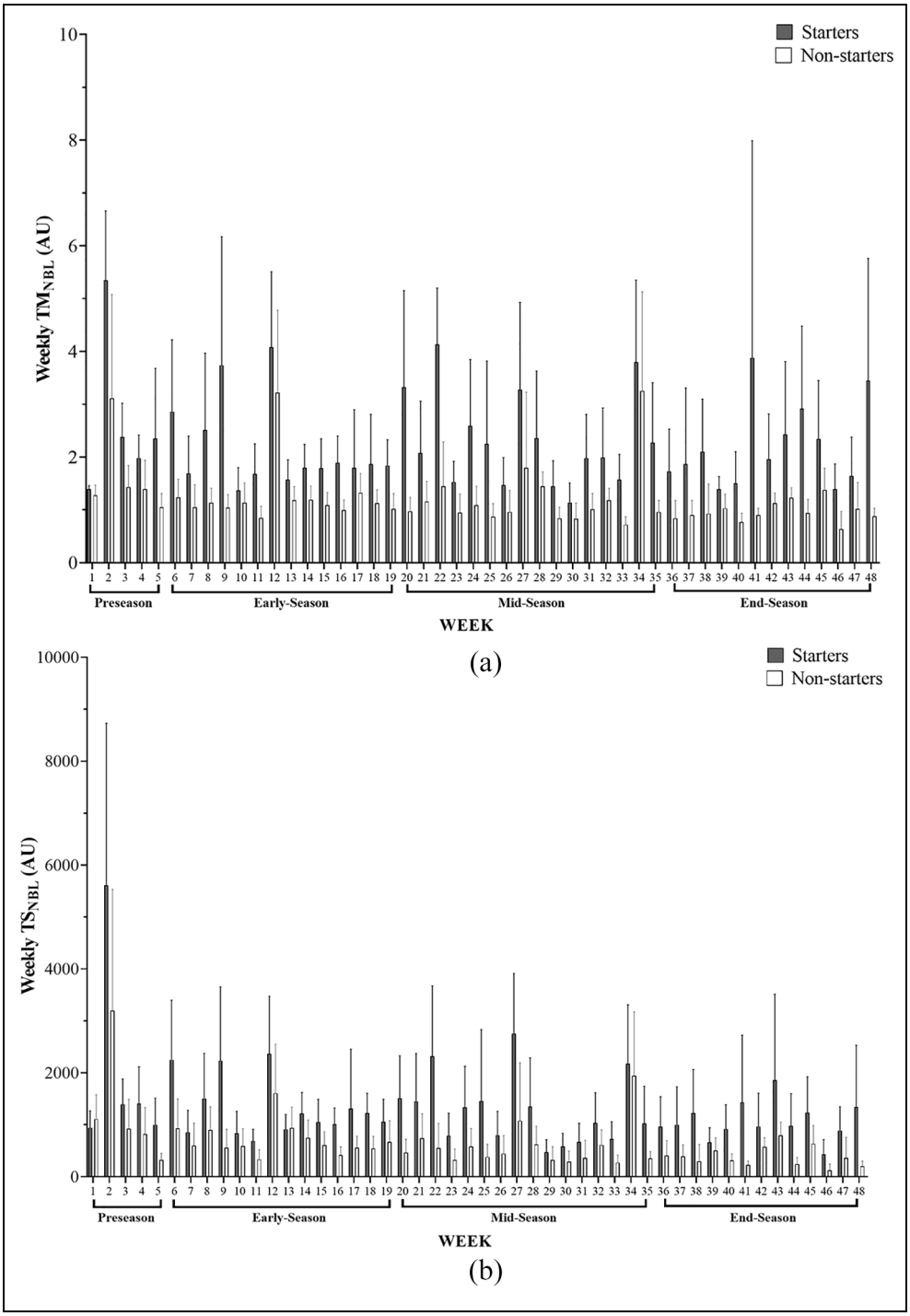

Figure 1. Descriptive statistics of weekly average for training: (a) monotony and (b) strain and their variations across the full season based on new body load.
Weekly TM_{NBL}: weekly training monotony based on new body load; Weekly TS_{NBL}: weekly training strain based on new body load.

starters could be the reason for this result; this possibility demands further investigation.

Regarding wTM_{NBL}, starters presented higher values than non-starters for the whole season, and the most prominent differences were reported in the mid-season. It has been suggested that training monotony values higher than 2 AU increase injury risk.[19] The current values are higher than those presented in previous research that has evaluated the training monotony based on players' rate of perceived exertion.[19,28] At this point, wTM_{NBL} and wTM_{Mba} may present a different characteristic than the previously adopted RPE-based measure. Therefore, future studies should aim to verify the relationship between NBL-based, MPA-based training monotony, and injury risky to establish reference values.

Additionally, the between-groups comparison revealed large differences throughout the season, with

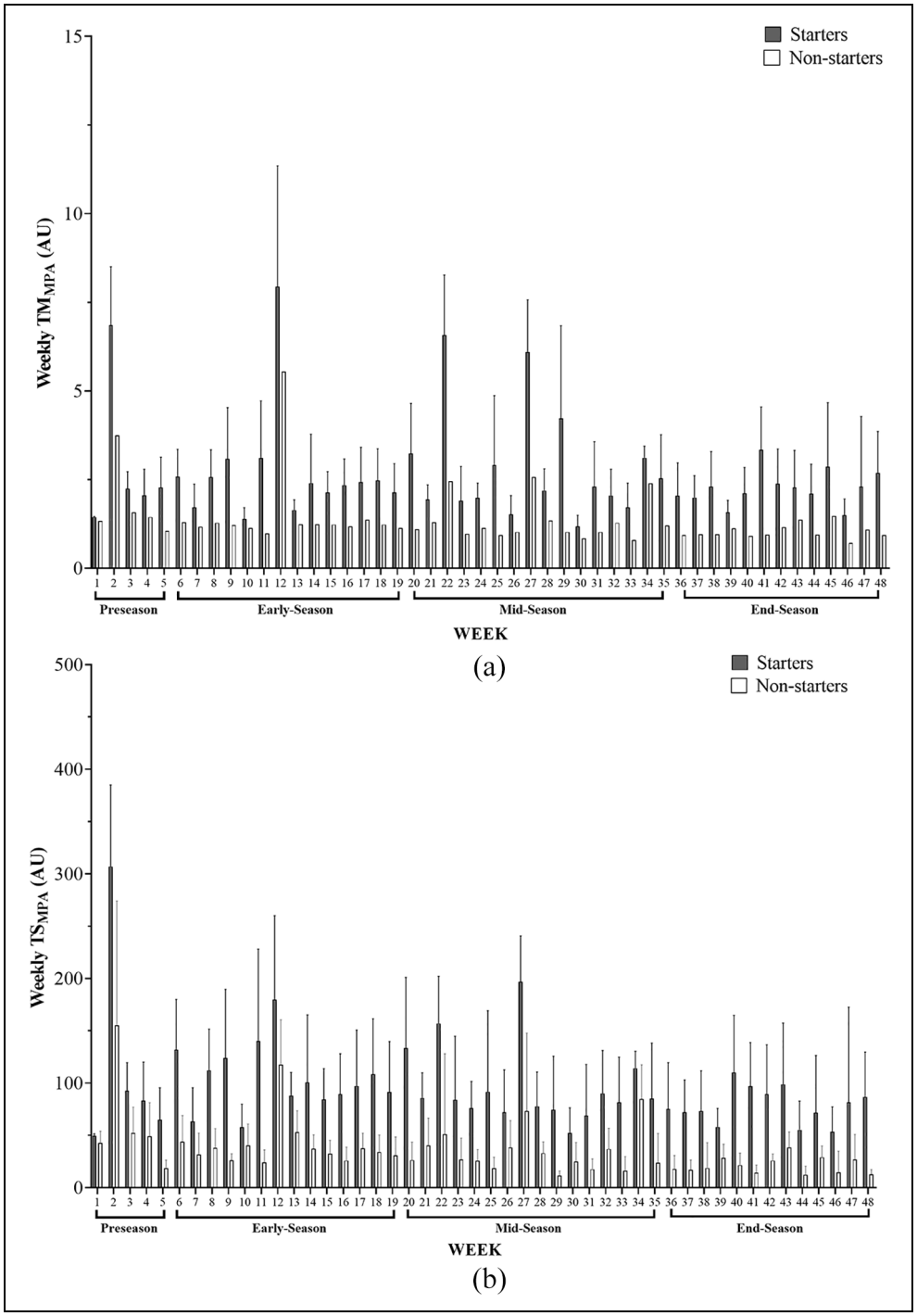

Figure 2. Descriptive statistics of weekly average for training: (a) monotony and (b) strain and their variations across the full season based on metabolic power average.
Weekly TM_{MPA}: weekly training monotony based on metabolic power average; Weekly TS_{MPA}: weekly training strain based on metabolic power average.

starters presenting higher values for every measure. The high-intensity activity demand required in official matches, such as acceleration and high-speed running,[2,10] might explain these between-groups differences during the in-season period. However, preseason differences were also reported. Similar to the rationale presented regarding wTS_{NBL} and wTS_{MPA}, some individual characteristics of the players may have impacted their external workloads during training sessions. Future studies should address this issue.

The data provided by this study illustrates the importance of constantly monitoring the players' workload. Values of TM higher than 2.0 AU have been cited as a risk factor for overtraining and injuries in players.[19] In the present study, values higher than 2.0 were reported during many weeks, but for different reasons.

Table 2. Differences between starters and non-starters on training monotony and strain based on new body load and metabolic power average in the different season periods.

	Season period	Group		% Difference (non-starters–starters)	*p*	Cohen's *d* (95% CI) (non-starters–starters)
		Starters	Non-starters			
TM_{NBL} (AU)	Preseason	2.69 (0.52)	1.57 (0.62)	−45.2 (−59.6 to −25.8)	< **0.001**	−1.87 (−2.89 to −0.84)
	Early-season	2.18 (0.49)	1.26 (0.20)	−41.7 (−51.4 to −29.9)	< **0.001**	−2.42 (−3.55 to −1.30)
	Mid-season	2.32 (0.46)	1.22 (0.25)	−47.6 (−56.7 to −36.5)	< **0.001**	−2.89 (−4.11 to −1.66)
	End-season	2.20 (0.58)	0.98 (0.12)	−54.4 (−63.1 to −43.6)	< **0.001**	−2.89 (−4.11 to −1.66)
TS_{NBL} (AU)	Preseason	2071.58 (654.62)	1156.49 (676.42)	−53.0 (−72.3 to −20.2)	**0.005**	−1.32 (−2.26 to −0.37)
	Early-season	1322.48 (226.85)	715.11 (199.08)	−47.4 (−58.3 to −33.5)	< **0.001**	−2.74 (−3.93 to −1.55)
	Mid-season	1278.37 (267.24)	587.65 (170.93)	−54.8 (−64.3 to −42.8)	< **0.001**	−2.99 (−4.23 to −1.74)
	End-season	1071.92 (291.63)	397.95 (87.00)	−62.2 (−70.5 to −51.6)	< **0.001**	−3.07 (−4.34 to −1.81)
TM_{MPA} (AU)	Preseason	2.97 (0.41)	1.72 (0.70)	−45.9 (−59.6 to −27.6)	< **0.001**	−2.07 (−3.13 to −1.01)
	Early-season	2.70 (0.56)	1.52 (0.18)	−43.1 (−52.2 to −32.2)	< **0.001**	−2.79 (−3.99 to −1.59)
	Mid-season	2.84 (0.46)	1.34 (0.42)	−54.1 (−62.9 to −43.2)	< **0.001**	−3.28 (−4.59 to −1.97)
	End-season	2.26 (0.51)	1.05 (0.15)	−52.9 (−60.8 to −43.4)	< **0.001**	−3.16 (−4.44 to −1.88)
TS_{MPA} (AU)	Preseason	119.43 (17.73)	58.27 (35.20)	−59.8 (−75.5 to −34.3)	< **0.001**	−2.07 (−3.14 to −1.01)
	Early-season	104.67 (25.47)	40.98 (8.95)	−60.5 (−68.7 to −50.2)	< **0.001**	−3.27 (−4.58 to −1.96)
	Mid-season	96.32 (19.38)	34.46 (16.09)	−66.4 (−75.3 to −54.4)	< **0.001**	−3.25 (−4.68 to −2.02)
	End-season	78.51 (21.14)	21.68 (6.21)	−72.4 (−78.8 to −63.9)	< **0.001**	−3.58 (−4.96 to −2.20)

Significant differences ($p \leqslant 0.05$) are highlighted in bold.

AU: arbitrary units; TM_{NBL}: weekly average training monotony based on new body load in AU; TS_{NBL}: weekly average training strain based on new body load in AU; TM_{MPA}: weekly average training monotony based on metabolic power average in AU; TS_{MPA}: weekly average training strain based on metabolic power average in AU; *p*: *p*-value at alpha level 0.05; Cohen's *d* (95% CI): Cohen's *d* effect size magnitude with 95% confidence interval.

To adequately prescribe training sessions, coaches must consider the values of both TM_{NBL} and TS_{NBL}. For example, in week 2, starters presented high values of $TM_{NBL,}$ followed by high values of TS_{NBL}. In this case, the growing monotony might be explained by a constant, but increased training workload. Therefore, there might be a risk of overtraining and injuries if similar patterns are repeated over the following weeks. On the other hand, similar to a previous study,[3] high TM_{NBL} values were also reported in week 41, although TS_{NBL} values were considered low. This result might indicate a constant low workload, which if chronically repeated, might lead to a reduction in the positive adaptations to the training.

The results of the current study have several practical applications. First, as previously recommended, coaches must consider simultaneously the values of TM and TS when analyzing the current workloads and defining the need for adjustments in training sessions. With starters presenting higher workloads than non-starters consistently over the season, specific training programs should be developed for each group. By developing specific programs, starters would likely have a reduced injury risk when their training workload is reduced (since their workload is usually higher) and non-starters would be exposed to stimuli enough to provide positive adaptations when their training workload is increased (as their workload is usually lower). The reported differences must be considered by strength and conditioning coaches since training programs that equally benefit both groups are important in soccer.[13]

To the best of the authors' knowledge, this is the first study to examine the training monotony and strain, measured by the NBL and MPA, throughout an entire soccer season. This study is also innovative in that it shows how the external workloads change based on players' statuses (i.e. starter or non-starter), which has not been addressed in the literature. In addition to its innovative contributions, some limitations must be addressed. The study is not large enough to make full generalizations. Furthermore, considering that this study took place in the real context of competition matches, controlling intervenient variables, such as motivation, competitive schedule, and environmental conditions, was not possible. Therefore, future studies should aim to investigate the variations of the training workload in different scenarios, such as different levels (like youth academies), clubs, and countries.

Conclusion

Starters regularly present higher values of NBL and MPA-based weekly training monotony and training strain than non-starters. This result must be taken into account when planning weekly workloads for these groups. Specifically, starters may experience high values of external workloads because of match-related demands. Therefore, weekly adjustments in the training workload are required to reduce injury risk. On the other hand, non-starters may experience a lack of high-intensity demands if they follow the same training regimen as starters. As such, the training workloads of these players should be increased to allow them to develop physical skills, thereby enabling them to achieve a high-level of performance when playing in official matches.

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